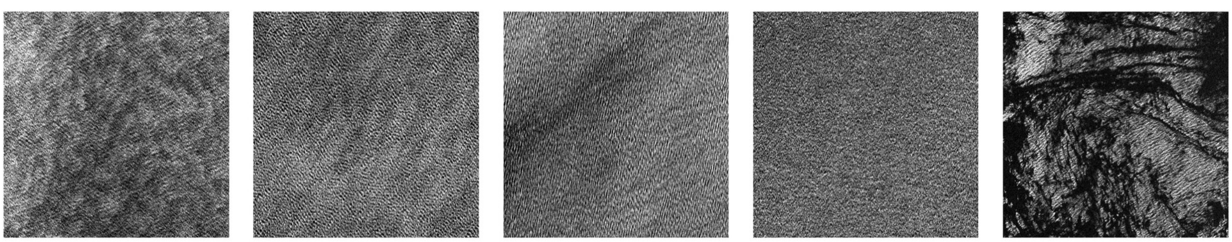
NeurIPS 2022: Can Three Million Satellite Radar Images Buy Their Way Out of the Labeling Bottleneck?

A University of Hawaiʻi team applies self-supervised contrastive learning to Sentinel-1 ocean radar, and reports honestly where it lands

The European Space Agency's Sentinel-1 radar satellites have one very convenient property: they image the ocean with a C-band synthetic aperture radar (SAR) rather than visible light, so they see through clouds, day or night. Between the two spacecraft, they collect nearly 120,000 "Wave mode" (WV) images of the open ocean every month, each covering a 20 km by 20 km patch at 5 m resolution. Those images encode a rich mix of atmospheric and ocean-surface phenomena, including wind, turbulence, ocean fronts, and biological slicks, making them a valuable record for studying air-sea interaction and boundary-layer dynamics.

The catch is that teaching a machine to read these images is hard. Deep learning needs labeled training data, and SAR vignettes can only be annotated by trained experts, so labeling more than a few thousand images quickly becomes infeasible. Sparse labels have been the main obstacle to putting this enormous image stream to work.

At the ML4PS workshop at NeurIPS 2022, Yannik Glaser, Peter Sadowski, and Justin E. Stopa of the University of Hawaiʻi at Mānoa asked a direct question: if labeled data is so scarce, can a model first teach itself useful representations from three million unlabeled Sentinel-1 images, and then be fine-tuned with the few labels that do exist? That is exactly the setting self-supervised contrastive learning is built for.



(a) MC (b) MC/WS (c) WS (d) NV (e) OT

Figure 1. Five representative SAR ocean scenes (each 20 × 20 km), ordered by atmospheric stability from unstable to stable: (a) strongly unstable micro-scale convective cells (MC); (b) mixed convection cells and wind streaks (MC/WS); (c) near-neutral wind streaks (WS); (d) stable, negligible variability (NV); and (e) a biological surface slick, labeled "other" (OT) in this study. Because these phenomena routinely coexist in a single scene, the authors switch to multi-label annotation.

Paper: https://ml4physicalsciences.github.io/2022/files/NeurIPS_ML4PS_2022_158.pdf

The problem: scarce labels, and a biased dataset on top

Earlier work by Stopa et al. classified SAR images into three surface-stability regimes, unstable, near-neutral, and stable, using the convolutional network of Wang et al. (referred to here as CmWV). That approach carried two structural flaws. First, its training set was cherry-picked: only exemplary images were kept for each class, so it did not reflect the true distribution of WV imagery. Second, the annotation protocol allowed only a single label per image, even though wind streaks, convective cells, and other phenomena often appear together in the same scene. The result was a training set that misrepresented the real ocean population.

The paper starts by rebuilding the benchmark on firmer ground: roughly 2,300 randomly sampled WV images, annotated by several experts into a consensus ground truth, with multiple labels allowed per image. The task focuses on four classes: micro-scale convective cells (MC), wind streaks (WS), negligible variability (NV), and other (OT). NV is defined as the absence of WS, MC, and any other phenomenon, while OT means no WS or MC but some other labeled phenomenon present. The dataset is split 60/20/20 into training, validation, and held-out test sets.

The method: learn a representation with SwAV, then probe it three ways

Contrastive self-supervised learning works by pulling different augmented views of the same image close together in representation space while pushing different images apart, all without human labels. The authors use SwAV (Swapping Assignments between multiple Views). Its neat trick is a clustering pretext task: representations are assigned to cluster centroids, and the network learns to predict one view's cluster assignment from another view. Where SimCLR needs prohibitively large batches, SwAV stores cluster assignments from past batches in a queue, so it can train at modest batch sizes and stays friendlier to limited compute.

A standard ResNet-50 backbone is trained with SwAV at batch size 1024 across 8 NVIDIA V100 32GB GPUs, with a queue of 16 batches and 1,000 cluster centroids, stopped after 65 epochs and about 10 days because of time and compute limits. Fine-tuning itself runs on a single V100 GPU at batch size 128. Once the unsupervised representation exists, how good is it? The authors answer with three downstream protocols of increasing depth, chosen precisely to separate the value of pretraining from the value of fine-tuning.

First, a weighted k-nearest-neighbor (kNN) classifier votes directly on the embeddings by Euclidean distance. With only one hyperparameter (k), it is a clean probe of representation quality. Second, linear evaluation freezes the entire ResNet and trains a single softmax layer on top of the frozen features. Third, end-to-end fine-tuning unfreezes all weights and trains the whole network on the labeled data. The last two protocols are run twice, once from ImageNet weights and once from SwAV weights, so the two initializations can be compared head to head.

The results: both crush the old model, but self-supervision doesn't beat ImageNet

There are two ways to read the numbers. Start with the representation itself. In the two protocols that keep the backbone frozen (kNN and linear evaluation), the self-supervised SwAV weights do hold a small edge over ImageNet: 0.864 versus 0.859 under kNN, and 0.841 versus

0.836 under linear evaluation. When the backbone is frozen, a representation learned on millions of SAR images carries a slight advantage.

Table 1(a). Micro-averaged AUROC by evaluation protocol: kNN, Linear (linear evaluation), and Fine-tune (end-to-end). With the backbone frozen (kNN, Linear) the SwAV weights edge ahead; once the network is fine-tuned end to end, the ImageNet weights are slightly higher. Every gap is small.

Evaluation protocol	Contrastive (SwAV) weights	ImageNet weights
kNN	0.864	0.859
Linear	0.841	0.836
Fine-tune	0.929	0.931

But the model you actually deploy is the fine-tuned one, and here the comparison reverses. Once all weights are unfrozen and trained end to end, the ImageNet initialization comes out slightly ahead (0.931 versus 0.929), with a best micro-averaged AUROC of about 0.93. In other words, the small advantage from self-supervised pretraining disappears once the whole network is free to adapt.

Against the previous state of the art, though, the gains are real. On the two hardest atmospheric classes, both new models clearly beat CmWV regardless of initialization: wind streaks (WS) rise from CmWV's 0.727 to 0.831–0.850, and convective cells (MC) from 0.793 to 0.872–0.873. On stable, featureless ocean (NV) all three models are essentially tied, around 0.95. Because the label definitions and classes differ between the old and new datasets, the OT class cannot be compared with CmWV at all.

Table 1(b). Per-class micro-averaged AUROC for the fine-tuned models. On the toughest classes, WS and MC, both new models far exceed the older CmWV; on NV the three are close; OT cannot be compared with CmWV because of differing label criteria (shown as –).

Model	MC	WS	NV	OT
SwAV weights	0.872	0.831	0.952	0.910
ImageNet weights	0.873	0.850	0.960	0.905
CmWV	0.793	0.727	0.946	–

The takeaway: an honest first look

The most commendable thing about this paper is its honesty. The authors do not dress the result up as a win for self-supervision. They state plainly that in this initial study, contrastive learning offers only marginal gains over the far cheaper standard practice of transferring features from a model trained on natural images, while costing substantially more compute.

That is not a dismissal of the idea. The authors stress these are preliminary results: longer training, more extensive hyperparameter tuning, and pretext tasks designed specifically for remote sensing all remain unexplored, and any of them could be where self-supervision finally pays off. For a domain where annotation requires experts but unlabeled imagery is abundant, methods that can exploit that unlabeled stream remain genuinely valuable.

For downstream applications, there is a more immediate payoff. Whether you go the self-supervised route or the ImageNet-transfer route, detection accuracy jumps sharply over the previous state-of-the-art classifier. For scientific and environmental tasks that rely on SAR imagery to estimate ocean-surface stability automatically, a more reliable model means a more trustworthy pipeline, and that benefit is available today.